

# **Threat Intelligence Driven Cybersecurity Risk Management Tool**

## Table of Contents

|                                                |           |
|------------------------------------------------|-----------|
| <b>Abstract.....</b>                           | <b>3</b>  |
| <b>Introduction.....</b>                       | <b>3</b>  |
| Background .....                               | 3         |
| Objectives .....                               | 3         |
| Scope.....                                     | 3         |
| <b>System Overview .....</b>                   | <b>5</b>  |
| <b>Architecture Diagram .....</b>              | <b>6</b>  |
| <b>Methodology.....</b>                        | <b>7</b>  |
| Step 1: Data Collection.....                   | 7         |
| Step 2: Data Processing.....                   | 7         |
| Step 3: Mapping Module .....                   | 7         |
| Step 4: Prediction Engine.....                 | 8         |
| Step 5: Visualization and User Interface ..... | 8         |
| <b>Implementation .....</b>                    | <b>8</b>  |
| 1. Data Retrieval and Preprocessing.....       | 8         |
| 2. Mapping Vulnerabilities to Techniques ..... | 9         |
| 3. Prediction Engine .....                     | 9         |
| 4. User Interface .....                        | 10        |
| 5. Visualization .....                         | 11        |
| <b>Results .....</b>                           | <b>12</b> |
| <b>Challenges and Limitations .....</b>        | <b>14</b> |
| <b>Conclusion and Future Work.....</b>         | <b>15</b> |
| Conclusion .....                               | 15        |
| Future Work.....                               | 15        |
| <b>References .....</b>                        | <b>16</b> |

## Abstract

Cybersecurity is a growing concern as attacks on systems and data become more common. This project aims to make it easier to understand how attackers operate and connect their techniques to known software vulnerabilities. Using tools like the MITRE ATT&CK framework and data from the National Vulnerability Database (NVD), the project creates a platform to predict threats and help organizations protect their systems more effectively. By linking attack patterns to vulnerabilities, this system provides clear insights and helps focus security efforts where they are needed most.

## Introduction

### Background

Cyberattacks are increasing every year, often because organizations fail to connect the way attackers work (their methods) with weaknesses in their systems. Without this connection, it's harder to prepare for or stop attacks. For example, while we know certain hackers use specific techniques, figuring out which systems or software they might target is not always clear.

### Objectives

This project aims to solve this problem by creating a tool that links attacker techniques to known vulnerabilities. With this tool, organizations can:

1. Predict possible attackers based on existing system weaknesses.
2. Focus their defences on the most critical vulnerabilities.
3. Save time and resources by having a clearer understanding of potential threats.

### Scope

The system uses widely available data, such as known vulnerabilities (CVE data) and attacker techniques (TTPs), to provide:

- A clear connection between threats and system weaknesses.
- Easy-to-understand visuals and insights for decision-makers.
- A user-friendly interface for security teams to work with.



# System Overview

The project integrates several components to create a comprehensive threat intelligence platform. It combines real-time data retrieval, machine learning, and a graphical interface to provide insights into potential cybersecurity threats. Below is a summary of the system's core elements:

## Key Components

### 1. Data Sources:

- CVE (Common Vulnerabilities and Exposures) data from NVD (National Vulnerability Database).
- TTP (Tactics, Techniques, and Procedures) data from the MITRE ATT&CK framework.
- Threat feeds from Open Threat Exchange (OTX).

### 2. Data Processing Modules:

- Scripts to fetch and preprocess CVE and TTP data.
- Algorithms for mapping vulnerabilities to attacker techniques.

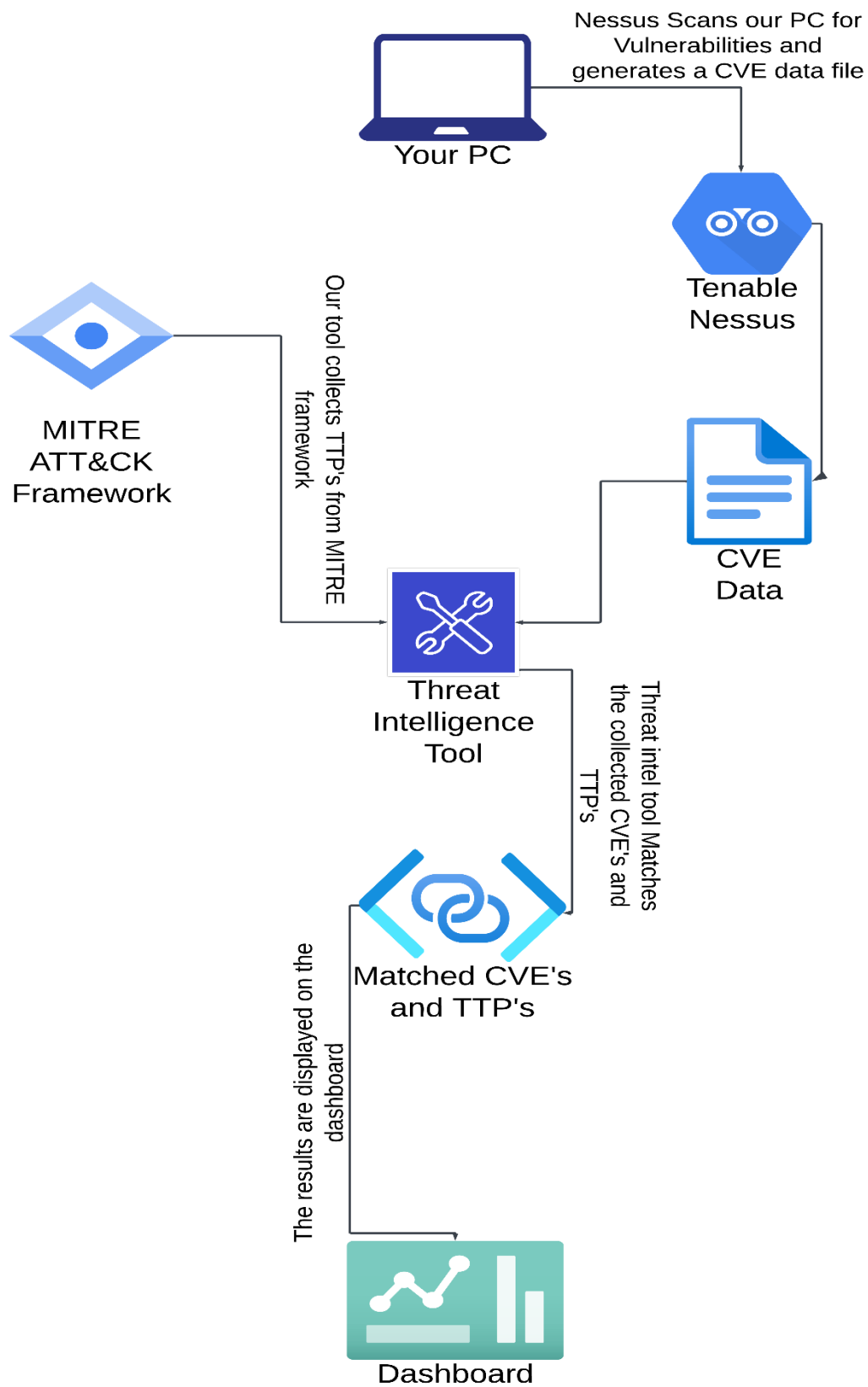
### 3. Prediction Engine:

- Uses data mapping to identify potential attackers based on known vulnerabilities.
- Outputs a list of predicted adversaries and associated techniques.

### 4. User Interface:

- A dashboard where users can view predictions, analysis, and threat mappings.
- Interactive features like graphs and tables for exploring data.

## Architecture Diagram



# Methodology

## Step 1: Data Collection

- Sources:
  - ❖ CVE data from the National Vulnerability Database (NVD) using the `get_cve_details.py` script.
  - ❖ TTP data from the MITRE ATT&CK framework using the `get_mitre_ttp.py` script.
  - ❖ Threat feeds from the Open Threat Exchange (OTX) via the `threatfeed.py` script.
- Purpose: Collect comprehensive information on vulnerabilities, attack techniques, and real-time threat intelligence to build a unified dataset.

## Step 2: Data Processing

- **Preprocessing:**
  - ❖ CVE data is cleaned and formatted using keyword matching to enhance descriptions.
  - ❖ TTP data is structured to include relevant attack patterns, descriptions, and platforms.
  - ❖ Threat feed data is filtered for high-relevance pulses to ensure quality.
  - ❖ `custom_cve_mapping.py` matches CVE descriptions with TTP data using a keyword correlation algorithm.

## Step 3: Mapping Module

- **Objective:** Create a link between known vulnerabilities (CVE) and adversary behaviours (TTP).
- **Process:**
  - ❖ Use the `adversary_ttp_mapping.py` script to connect adversary profiles to TTPs.
  - ❖ Generate a comprehensive CSV file listing mapped adversaries, techniques, and vulnerabilities.

## Step 4: Prediction Engine

- **Goal:** Predict the most likely attackers based on existing vulnerabilities.
- **Logic:**
  - ❖ Analyse attack patterns to determine adversary likelihoods.
  - ❖ Use the `predict_adv.py` script to generate predictions based on mapped data.
- **Output:**
  - ❖ CSV files summarizing predicted adversaries and their respective TTP counts.

## Step 5: Visualization and User Interface

- **Dashboard:**
  - ❖ Developed using Python's Tkinter library, the GUI enables users to interact with the data and visual outputs.
  - ❖ Key Features:
    - Display of top adversaries with bar charts (via `predict_adv_gui.py`).
    - Tables summarizing CVE-TTP mappings and threat insights.
  - ❖ **Interactivity:** Users can fetch threat data, view mappings, and explore details of CVEs and adversaries.

# Implementation

## 1. Data Retrieval and Preprocessing

- **Scripts Used:**
  - `get_cve_details.py`: Fetches CVE details from the NVD database, extracting key attributes like descriptions, severity scores, and references.
  - `get_mitre_ttp.py`: Retrieves TTP data from the MITRE ATT&CK framework, focusing on attack patterns and their descriptions.



- threatfeed.py: Pulls real-time threat intelligence pulses from OTX to incorporate recent attacks into the dataset.

```
1 def fetch_cve_data(cve_id):
2     url = f"https://services.nvd.nist.gov/rest/json/cves/2.0?cveId={cve_id}"
3     headers = {'apiKey': NVD_API_KEY}
4     response = requests.get(url, headers=headers)
5     return response.json() if response.status_code == 200 else None
```

## 2. Mapping Vulnerabilities to Techniques

- **Purpose:** Establish links between CVEs and TTPs using keyword correlation.
- **Scripts Used:**
  - custom\_cve\_mapping.py: Matches CVE descriptions with TTP descriptions to identify overlaps.
  - adversary\_ttp\_mapping.py: Maps adversary groups to techniques using the MITRE ATT&CK API.

### Code Snippet: Mapping CVEs to TTPs

```
1 def correlate_cve_ttp_by_keywords(cve_df, ttp_df, keywords):
2     correlations = []
3     for _, cve_row in cve_df.iterrows():
4         for _, ttp_row in ttp_df.iterrows():
5             if any(keyword in cve_row['Description'] and ttp_row['Description'] for keyword in keywords):
6                 correlations.append({
7                     'CVE ID': cve_row['CVE ID'],
8                     'TTP ID': ttp_row['TTP'],
9                     'TTP Name': ttp_row['Name']
10                })
11     return pd.DataFrame(correlations)
```

## 3. Prediction Engine

- **Functionality:**
  - The engine predicts potential attackers based on mappings of CVEs to TTPs.
  - Outputs ranked adversaries based on their likelihood of exploiting identified vulnerabilities.

### Code Snippet: Prediction Logic

```

1 def predict_adversaries(output_csv='predicted_adversaries.csv'):
2     adversary_data = []
3     for _, row in cve_ttp_df.iterrows():
4         if row['TTP'] in ttp_adversary_map:
5             for adversary in ttp_adversary_map[row['TTP']]:
6                 adversary_data.append({'TTP': row['TTP'], 'Adversary': adversary})
7     pd.DataFrame(adversary_data).to_csv(output_csv, index=False)

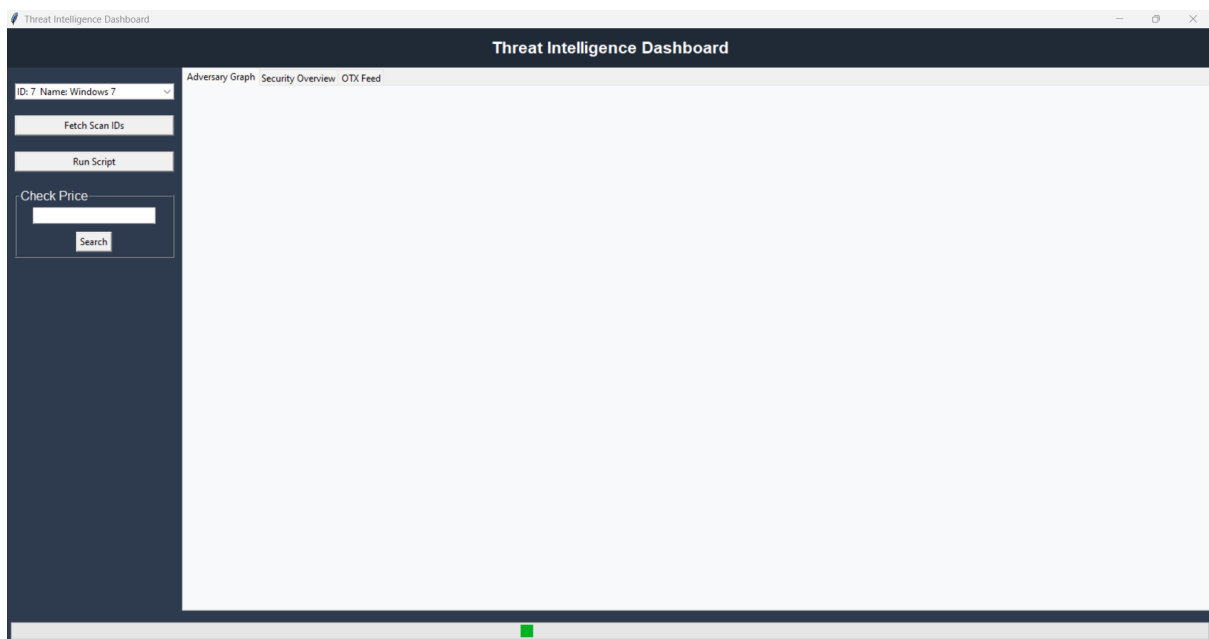
```

## 4. User Interface

- **GUI Design:**

- The GUI.py script creates an interactive dashboard for users to view predictions and mappings.
- Features include dropdown menus to fetch scan data, tables displaying CVE details, and graphs summarizing adversary activities. We can also search the prices of the solutions listed along with the links to purchase.

### Screenshot: Dashboard Overview



### Code Snippet: GUI Event Handling

```

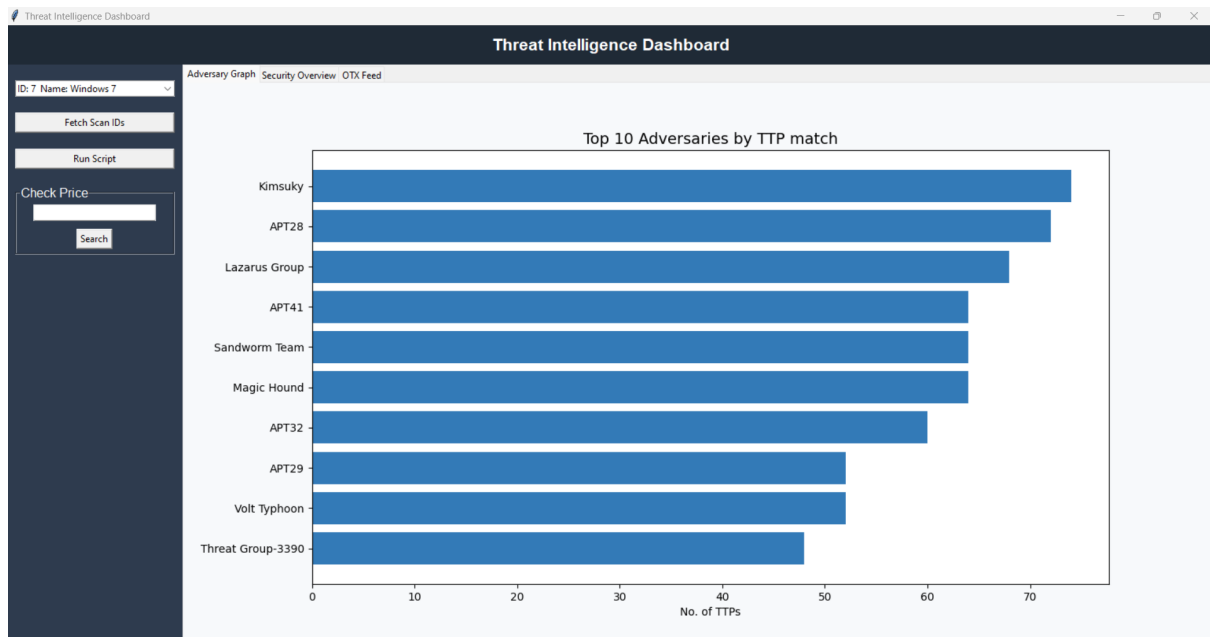
1 def execute_functions(scan_id):
2     process_scan_data(scan_id)
3     map_adversaries_ttps()
4     fetch_mitre_ttp_data()
5     map_cve()
6     predict_adversaries()
7     display_adversary_graph()
8     display_cve_table()

```

## 5. Visualization

- **Graphs:**
  - Bar charts show the most active adversaries based on their TTP counts.
  - Graphs are dynamically updated based on processed data.

### Screenshot: Adversary Bar Chart



# Results

## 1. CVE to TTP Mapping

The system successfully mapped known vulnerabilities (CVEs) to attacker techniques (TTPs) using keyword correlations and data from the MITRE ATT&CK framework.

## 2. Predicted Adversaries

The prediction engine identified adversaries based on their use of mapped TTPs, providing a ranked list for prioritization.

## 3. Real-Time Threat Insights

Using OTX threat feeds, the system provided updated information on emerging threats. These feeds were integrated into the dashboard for quick access to actionable intelligence.

### Sample Output: OTX Threat Feed

The screenshot displays the Threat Intelligence Dashboard. On the left, there's a sidebar with a dropdown menu showing 'ID: 7 Name: Windows 7', buttons for 'Fetch Scan IDs', 'Run Script', and a 'Check Price' section with a search bar. The main area features a table titled 'Threat Intelligence Dashboard' with tabs for 'Adversary Graph', 'Security Overview', and 'OTX Feed'. The 'OTX Feed' tab is active, showing a table with columns: ID, Name, Description, Created, and References. Below the table, a modal window titled 'OTX Pulse Details' is open, showing details for a specific threat feed entry.

| ID                       | Name                                                               | Description                                     | Created                    | References                                                                                                                                                |
|--------------------------|--------------------------------------------------------------------|-------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6729af15e31c9eab16197899 | Automatically Detecting DNS Hijacking in Passive DNS               | This article describes a machine learning-based | 2024-11-05T05:37:25.249000 | <a href="https://unit42.paloaltonetworks.com/detect-dns-hijacking-passive-dns/">https://unit42.paloaltonetworks.com/detect-dns-hijacking-passive-dns/</a> |
| 672df7e31f1576bf7478bf77 | QSC: new modular framework in CloudComputing campaigns             | Kaspersky researchers discovered QSC, a multi-j | 2024-11-08T11:37:07.686000 | <a href="https://securelist.com/cloudcomputing-qsc-framework/114438">https://securelist.com/cloudcomputing-qsc-framework/114438</a>                       |
| 673937f9370f10b3cab732eb | Telekopye transitions to targeting tourists via hotel booking scam | ESET researchers have discovered that Telekopye | 2024-11-17T00:25:35.270000 | <a href="https://www.welivesecurity.com/en/eset-research/telekopye-hits">https://www.welivesecurity.com/en/eset-research/telekopye-hits</a>               |
| 673d0a2938340b4f61b0f3a  | Zoom-In: A Closer Look into the Malware Artifacts, Behaviors and   | FrostyGoop, an operational technology (OT) m    | 2024-11-19T21:59:05.639000 | <a href="https://unit42.paloaltonetworks.com/frostygoop-malware-analysis">https://unit42.paloaltonetworks.com/frostygoop-malware-analysis</a>             |
| 674736a47f4929a39b28fb6  | Gaming Engines: An Undetected Playground for Malware Loaders       | Check Point Research uncovered a new techniq    | 2024-11-27T15:11:32.952000 | <a href="https://research.checkpoint.com/2024/gaming-engines-an-unde">https://research.checkpoint.com/2024/gaming-engines-an-unde</a>                     |

**OTX Pulse Details**

ID: 6729af15e31c9eab16197899  
Name: Automatically Detecting DNS Hijacking in Passive DNS  
Description: This article describes a machine learning-based pipeline for detecting DNS hijacking using passive DNS data. The system processes an average of 167 million new DNS records daily, extracting 74 features from over 169 terabytes of data. Between March and September 2024, it identified 6,729 hijacking incidents out of 29 billion processed records. Notable examples include the hijacking of a Hungarian political party's domain, the defacement of a utility company and ISP, and the use of university and research center domains for illicit gambling. The pipeline can now detect DNS hijacking in customer traffic within 10 minutes, providing crucial protection against this pervasive threat.  
Created: 2024-11-05T05:37:25.249000  
References: <https://unit42.paloaltonetworks.com/detect-dns-hijacking-passive-dns/>

## 4. Dashboard Features

The interactive dashboard allowed users to:

- View mapped CVE-TTP data in tabular form.
- Access adversary predictions and their likelihood scores.
- Explore real-time threat intelligence feeds.

Threat Intelligence Dashboard

ID: 7 Name: Windows 7

Fetch Scan IDs

Run Script

Check Price

Search

Adversary Graph Security Overview OTX Feed

| Host          | CVE            | CVSS v2.0 Base Score | Protocol | Port | Description                                                                                                                                 | Solution                                                          |
|---------------|----------------|----------------------|----------|------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 192.168.1.199 | nan            | 6.4                  | tcp      | 8834 | The server's X.509 certificate cannot be trusted. This situation can occur in three different ways:                                         | Purchase or generate a proper SSL certificate for this service.   |
| 192.168.1.199 | CVE-2023-52323 | 5.4                  | tcp      | 0    | The version of PyCryptodome installed on the remote host is prior to 3.19.1. It is, therefore, affected by this vulnerability.              | Upgrade to PyCryptodome or PyCryptodomex version 3.19.1 or later. |
| 192.168.1.199 | CVE-2023-46809 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-21890 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-21891 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-21892 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-21896 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-22017 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-22019 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.19.1, 20.11.1, 21.6.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.19.1 / 20.11.1 / 21.6.2 or later.   |
| 192.168.1.199 | CVE-2024-27962 | 5.4                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.1, 20.12.1, 21.7.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.1 / 20.12.1 / 21.7.2 or later.   |
| 192.168.1.199 | CVE-2024-27963 | 5.4                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.1, 20.12.1, 21.7.2. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.1 / 20.12.1 / 21.7.2 or later.   |
| 192.168.1.199 | CVE-2024-27306 | 4.3                  | tcp      | 0    | The version of aiohttp installed on the remote host is prior to 3.9.4. It is, therefore, affected by this vulnerability.                    | Upgrade to aiohttp version 3.9.4 or later.                        |
| 192.168.1.199 | CVE-2024-37891 | 4.6                  | tcp      | 0    | urllib3 is a user-friendly HTTP client library for Python. When using urllib3's proxy support with                                          | Upgrade to urllib3 version 1.26.19, 2.2.2 or later.               |
| 192.168.1.199 | CVE-2024-22018 | 9.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.4, 20.15.1, 22.4.1. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.4 / 20.15.1 / 22.4.1 or later.   |
| 192.168.1.199 | CVE-2024-22020 | 9.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.4, 20.15.1, 22.4.1. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.4 / 20.15.1 / 22.4.1 or later.   |
| 192.168.1.199 | CVE-2024-27003 | 5.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.4, 20.15.1, 22.4.1. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.4 / 20.15.1 / 22.4.1 or later.   |
| 192.168.1.199 | CVE-2024-36137 | 9.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.4, 20.15.1, 22.4.1. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.4 / 20.15.1 / 22.4.1 or later.   |
| 192.168.1.199 | CVE-2024-37372 | 9.0                  | tcp      | 0    | The version of Node.js installed on the remote host is prior to 18.20.4, 20.15.1, 22.4.1. It is, therefore, affected by this vulnerability. | Upgrade to Node.js version 18.20.4 / 20.15.1 / 22.4.1 or later.   |
| 192.168.1.199 | CVE-2024-39689 | 7.8                  | tcp      | 0    | The detected version of Certifi python package, certifi, is prior to version 2024.07.04. It is, therefore, affected by this vulnerability.  | Upgrade to certifi version 2024.07.04 or later.                   |
| 192.168.1.199 | CVE-2024-45320 | 7.8                  | tcp      | 0    | The detected version of the Django Python package, Django, is 4.2.x prior to 4.2.16, 5.0.x prior to 5.0.9, 5.1.x prior to 5.1.1 or later.   | Upgrade to Django version 4.2.16, 5.0.9, 5.1.1 or later.          |
| 192.168.1.199 | CVE-2024-45321 | 7.8                  | tcp      | 0    | The detected version of the Django Python package, Django, is 4.2.x prior to 4.2.16, 5.0.x prior to 5.0.9, 5.1.x prior to 5.1.1 or later.   | Upgrade to Django version 4.2.16, 5.0.9, 5.1.1 or later.          |

# Challenges and Limitations

Despite the success of this project in linking vulnerabilities to attacker techniques and providing predictive insights, several challenges and limitations were encountered:

## 1. Data Availability and Quality

- **Issue:** Inconsistent or incomplete data from external sources like CVE databases or threat intelligence feeds.
- **Impact:** Gaps in the data reduced the effectiveness of the mapping and prediction algorithms.
- **Example:** Some CVE entries lacked detailed descriptions or severity scores, making it harder to perform accurate correlations.

## 2. Mapping Accuracy

- **Issue:** The process of linking CVEs to TTPs relies heavily on keyword matching.
- **Impact:** While effective for some cases, this approach can miss intricate connections or produce false positives.

## 3. Real-Time Integration

- **Issue:** Real-time updates from sources like OTX were limited to periodic fetches.
- **Impact:** This created a slight lag in providing the most current threat intelligence.

## 4. Predictive Model Scope

- **Issue:** The prediction engine relies on existing data, which may not account for Zero-day exploits.
- **Impact:** Newly discovered threats might not be effectively identified by the system.

# Conclusion and Future Work

## Conclusion

This project successfully demonstrated a system for connecting known vulnerabilities (CVEs) to adversarial techniques (TTPs) and predicting likely attackers. By using the MITRE ATT&CK framework, CVE databases, and real-time threat feeds, the system provides cybersecurity teams with actionable insights to prioritize defences and mitigate risks. The inclusion of a user-friendly dashboard further ensures that the results are accessible and practical for use in real-world scenarios.

## Future Work

### 1. Enhancing Data Mapping:

- Incorporate additional datasets to improve mappings, such as exploit databases and dark web intelligence.

### 2. Scalability:

- Optimize the system for larger datasets to handle enterprise-level threat intelligence requirements.

# References

## 1. CVE Details:

- National Vulnerability Database (NVD): <https://nvd.nist.gov/>

## 2. MITRE ATT&CK Framework:

- MITRE ATT&CK API: <https://attack.mitre.org/>

## 3. Open Threat Exchange (OTX):

- AlienVault OTX Platform: <https://otx.alienvault.com/>

## 4. Scripts and Tools:

- Python Libraries:
  - taxii2client for interacting with MITRE's API: <https://github.com/oasis-open/cti-taxii-client>
  - requests for HTTP requests: <https://docs.python-requests.org/>
  - pandas for data manipulation: <https://pandas.pydata.org/>
  - matplotlib for visualizations: <https://matplotlib.org/>

## 5. System Design and Implementation:

- Custom Scripts:
  - CVE Fetching: `get_cve_details.py`
  - TTP Fetching: `get_mitre_ttp.py`
  - Adversary Mapping: `adversary_ttp_mapping.py`
  - Prediction Engine: `predict_adv.py`

## 6. GUI Framework:

- Tkinter for GUI development: <https://docs.python.org/3/library/tkinter.html>